

NISHITA GUPTA

+91-9818604429 | nishitagupta1408@gmail.com | <https://www.linkedin.com/in/iamnishitagupta/> | github.com/Nishita-1408

EDUCATION

Indian Institute of Technology, Mandi

BTech in Mathematics and Computing

2023 – Present

CGPA: 8.86/10

Amity International School, Mayur Vihar

Senior Secondary Education

2022 - 2023

Percentage: 96%

EXPERIENCE

Flipkart | *Software Development Engineer Intern*

Jun 2026 – Jul 2026

- Migrated pricing logic to the new Helix platform by **adding derived value tags** (BUDGET_PICK and TOP_VALUE) with sale-context gating, **dynamically highlighting** budget-friendly items during BAU operations and top-value items during tier-1 sales.
- Resolved legacy framework incompatibilities by **utilizing Java reflection** to add Jakarta JAX-RS support, **safely exposing** a crucial filter trail across old and new systems using a single agent JAR.
- Streamlined complex offer debugging for engineers by **orchestrating a consolidated POST API** and MCP tool pipeline, **slashing manual request construction** from 50+ lines of JSON down to just 3 fields.
- Optimized chatbot response payloads through integrated MCP tool response trimming, **slashing LLM token consumption by 50%** (from ~24k to ~12k tokens) while **preserving a stable API response time** of 576 ms.

PROJECTS

Multi-Objective Evolutionary Portfolio Optimisation | *NSGA-III, MOEA/D, Python, Ledoit-Wolf, GARCH*

Jan 2026 – May 2026

- Engineered an extensive multi-objective evolutionary portfolio optimisation framework** across S&P 500 and LSE datasets (2015–2024), **implementing higher-moment MVSK objectives** (skewness and kurtosis) optimized via NSGA-III reference-direction niching.
- Integrated robust covariance estimation mechanics** leveraging Ledoit-Wolf shrinkage and Random Matrix Theory (RMT) de-noising to **mitigate high-dimensional estimation noise** and secure structural portfolio stability.
- Incorporated parametric Student's t-distribution CVaR** via moment matching, achieving a **peak Sharpe ratio of 1.055** with the Ledoit-MVSK NSGA-III model and proving heavy-tail risk optimization matches historical simulations.

Full-Stack Student Ride-Sharing App (Voyager) | *Flutter, Dart, Supabase, PostgreSQL, Firebase Auth, OSRM*

Aug 2025 – Dec 2025

- Architected a scalable dual-backend architecture** utilizing Firebase for secure user authentication and **Supabase (PostgreSQL) with Row Level Security (RLS)** to manage complex relational ride listings and digital ticket marketplaces.
- Engineered advanced geospatial intelligence features** using Nominatim geocoding and Open Source Routing Machine (OSRM) trip services, **implementing multi-stop pathing and path geometry filtering** to eliminate incorrect route matching.
- Developed a multi-party approval consensus system** requiring unanimous passenger voting workflows, complemented by **real-time data synchronization streams** and automated transaction logging.

TECHNICAL SKILLS

Languages: C++, Python, Java, MySQL

Technologies/Frameworks: Dropwizard, Flask, Kubernetes, Solidworks, Docker, Supabase, Git

RELEVANT COURSES

Mathematics: Ordinary Differential Equations, Linear Algebra, Probability and Statistics, Machine Learning, Mathematical Foundations of Computer Science, Partial Differential Equations, Applied Graph Theory, Mathematical Modelling

Computer Science: Introduction to Python and Data Science, Data Structures and Algorithms, Computer Organization, Formal Languages and Automata Theory, Applied Databases Practicum

ACHIEVEMENTS / EXTRACURRICULAR

- Solved **300+ questions** on [LeetCode](https://leetcode.com/).
- Secured an **All India Rank 21** out of 1.5 lakh+ candidates in NTSE.
- Semi-finalist in **Flipkart GRID 7.0** out of 1 lakh+ candidates.
- Secured international ranks of **250 (National Science Olympiad)** and **150 (International English Olympiad)** in SOF Olympiads among over 1 lakh global participants.
- Won Champion's Trophy, 1 Gold, and 1 Silver Medal for three consecutive years (2012–2014) in National Abacus Competitions organized by Brainobrain.
- Core member of Quosity (The Quizzing Club of IIT Mandi); Participated in Inter IIT Cult Meet (2023, 2024), won gold models in inter-college quizzes (General and Sci-Tech) and organized 15+ sessions.